UDCT v7.14 Exploratory Findings: The p=1.5 Boundary in QUMOND

A model-specific asymptotic threshold and finite-energy Osipkov–Merritt tangency audit

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Code repository: https://github.com/wspaik5/UDCT_v7.14_Exploratory_p1.5_Boundary

Abstract

This technical note consolidates the archived UDCT v7.14 exploratory Steps 01–50 and six subsequent numerical audits. In a specified isolated spherical simple-QUMOND Plummer-field construction, the tracer family rho_p(r)=[1+(r/a)^p]^(-5/p) has a central potential scaling Psi proportional to r^(3/2), implying a leading tracer-density expansion proportional to Psi^(2p/3). The coefficient of its second derivative changes sign across p=3/2 and vanishes at p=3/2. This asymptotic observation must not be conflated with a universal QUMOND threshold or an exact finite-energy distribution-function zero. Separate Osipkov–Merritt (OM) inversions show a model-specific local zero-DF tangency near ra/a=0.4664667874 at p=1.5. A six-point p scan, finite-domain sign checks, quadrature convergence, and two numerical optimizers support a smooth local critical curve, not a discontinuity at p=1.5. The calculations share a common physical inversion kernel and do not establish global DF positivity, dynamical stability, a 3D external-field result, or direct Hydrus I stellar likelihood agreement.

1. Scientific question and evidential scope

The motivating question is whether the central asymptotic exponent p=1.5 and the finite-energy OM anisotropy tangency represent one physical transition. The archived results instead require a two-boundary interpretation: an asymptotic coefficient cancellation and a continuously varying local tangency condition. The present document is a synthesis of existing exploratory artifacts, not a newly preregistered prediction. No post hoc numerical criterion is relabeled as confirmatory.

2. Source archives and provenance

The user-provided primary archives are “7.14 Hessian 1~25.zip” (early exploratory calculations), “7.14 Hessian 26~50.zip” (25 nested step packages), and “7.14 Exploratory audit 1~6.zip” (six nested audit packages). Audit 06 is separately available as an executable nine-file archive. A SHA-256 inventory is provided in the companion provenance manifest. A nested archive name is evidence of archival presence, not independent verification of every embedded result.

3. Defined model and normalization

The Step 40/Audit 01 model specifies G=4.30091e-3 pc (km/s)^2/Msun, M=32730 Msun, a=53.3 pc, a0=1.2e-10 m/s^2; a Plummer gravitating mass with Newtonian acceleration gN=G M r/(r^2+a^2)^(3/2); and the isolated spherical simple-QUMOND acceleration g=gN/2+sqrt(gN^2/4+a0 gN). The tracer density is rho_p=[1+(r/a)^p]^(-5/p), normalized to rho(0)=1. Potential and inversion conventions follow the archived scripts; numerical DF magnitudes are therefore convention- and normalization-dependent.

4. Derivation of the central asymptotic exponent

At r much smaller than a, gN is proportional to r. In the deep-acceleration central limit, the leading square-root term makes g proportional to r^(1/2). Integration from the center gives Psi(r)=Phi(r)-Phi(0) proportional to r^(3/2). Meanwhile rho_p(r)=1-(5/p)(r/a)^p+O(r^(2p)). Eliminating r gives rho_p(Psi)=1-C Psi^alpha+... with alpha=2p/3 and C positive under the stated conventions. Hence the leading second derivative is -C alpha(alpha-1) Psi^(alpha-2). Its coefficient vanishes at alpha=1, i.e. p=1.5, and the sign of this leading term reverses on crossing that exponent. At p=1.5 the omitted next-order terms determine the local behavior. Therefore the leading-term cancellation is not by itself a theorem that the full DF vanishes at all energies or that a global physical transition occurs.

5. Finite-energy OM inversion and local tangency

For the archived OM inversion, the numerical implementation uses an affine decomposition f(Q,ra)=f_iso(Q)+B(Q)/(ra/a)^2. When f_iso is positive and B negative, the local zero-DF condition gives (ra/a)^2=-B/f_iso. Maximizing this local critical-radius expression over a finite energy-label interval yields a candidate tangency. The energy-label r/a in these audits is an inversion-energy coordinate and should not be misread as an observed stellar orbital radius. This procedure neither certifies positivity over the full energy continuum nor proves that every distribution function is dynamically realizable.

6. Quantitative local critical curve

p	Critical ra/a (192 nodes)	Energy-label r/a
1.490	0.462739644607	0.476369776
1.495	0.464599468922	0.475243086
1.499	0.466092716172	0.474328250
1.500	0.466466787396	0.474097647
1.501	0.466841165090	0.473866280
1.505	0.468341767989	0.472933131

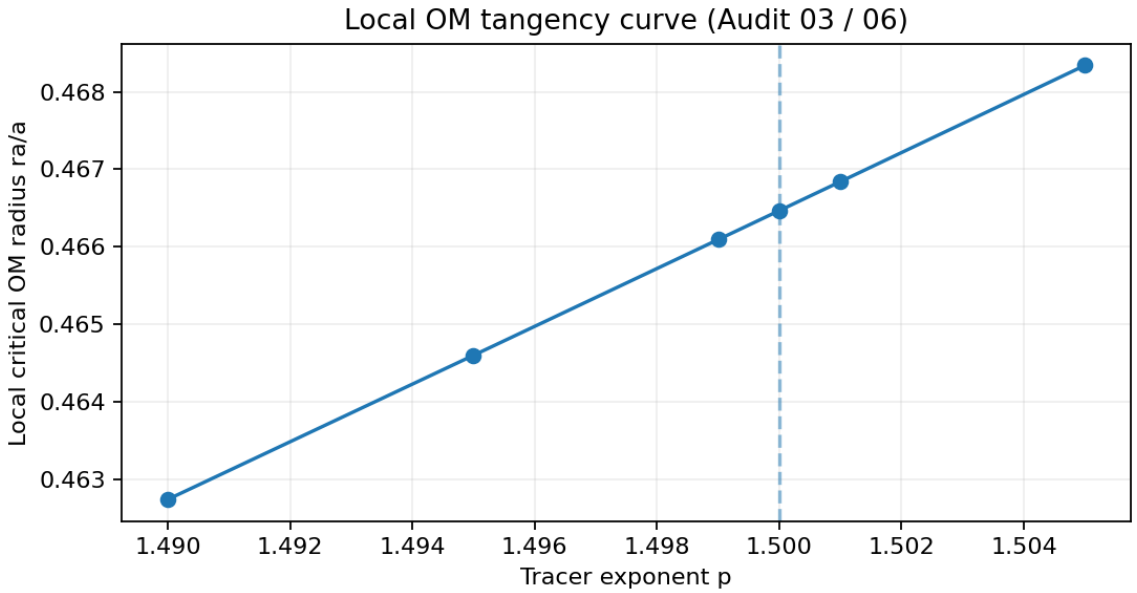


Figure 1. Local OM critical-radius curve from the archived Audit 03 input re-tabulated in Audit 06; this is not a global boundary.

7. Numerical cross-checks and convergence

Audit 04 evaluated 3 tracer exponents by 4 anisotropy settings by 18 energy labels (216 DF samples). At ra/a=0.45, seven sampled negative DF values were reported for each p=1.499, 1.500 and 1.501. At the p=1.5 reference radius ra/a=0.466466787396, p=1.501 yielded three negative samples whereas p=1.499 and 1.500 yielded none under the stated threshold f<-1e-10. These counts are finite-grid diagnostics, not global sign classifications.

Audit 05 fixed p=1.501 and ra/a=0.466466787396. The reported local minima were -3.2713868467e-6, -3.2713868390e-6 and -3.2713868336e-6 for 96, 192 and 384 Gauss–Legendre nodes, respectively. At 192 nodes the adjusted local zero-tangency radius was ra/a approximately 0.466841165090. This supports numerical persistence of a small negative lobe in that finite interval, but not exact error certification.

Audit 06 compared bounded minimization and golden-section search for p=1.499, 1.500 and 1.501; the maximum reported discrepancy in critical ra/a was 5.6e-17, reflecting agreement of optimization on the same numerical function. Central finite differences gave slope 0.37422446 and curvature 0.30646998 at h=0.001, versus slope 0.37422991 and curvature 0.30648475 at h=0.005. Their agreement supports local smoothness within the sampled model and numerical procedure, not analyticity or universal criticality.

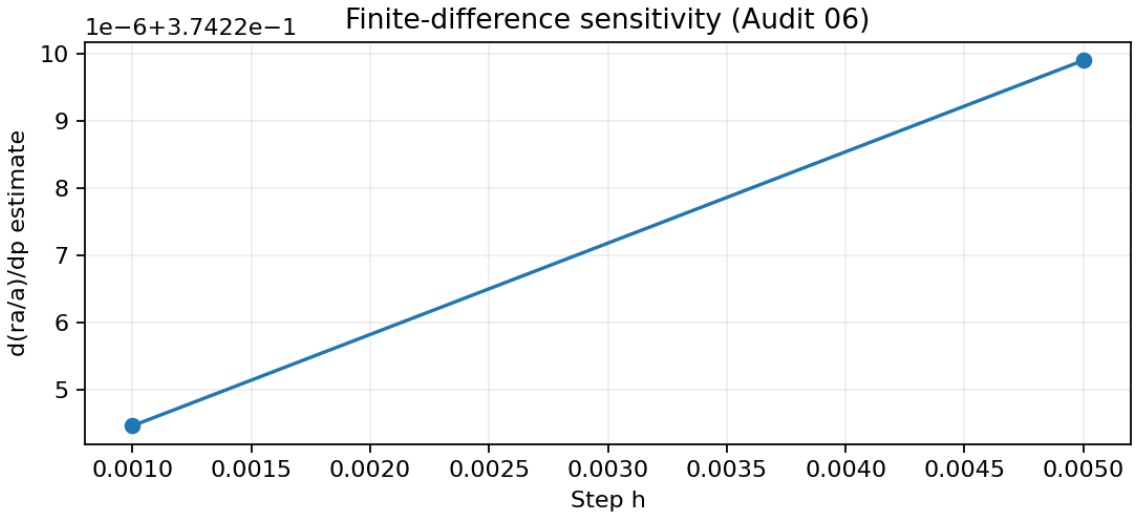


Figure 2. Finite-difference slope sensitivity at p=1.5; only two step sizes were archived.

8. Audit-by-audit evidence ledger

Audit	Question	What is supported
01	Asymptotics and independent quadrature	Leading exponent cancellation and numerical agreement; not a new inversion theorem
02	Finite energy scan	No resolved significant negative on the sampled reference tangency grid; not continuum positivity
03	Variation with p	Local critical ra/a changes continuously over sampled p values
04	Crossing p=1.5	Finite-grid sign differences distinguish asymptotic and OM conditions
05	Quadrature convergence	Persistent local negative lobe for p=1.501 at fixed reference anisotropy
06	Optimizer and derivative checks	Algorithm agreement and stable local finite-difference slope/curvature

9. Interpretation and falsifiability

The evidence supports two distinct mathematical features in the specified spherical model. The p=1.5 coefficient cancellation is an asymptotic statement; the OM local tangency is an energy- and anisotropy-dependent numerical

condition. The observed smooth movement of the latter across p=1.5 is incompatible with treating p=1.5 as an automatically singular finite-energy OM radius. A later preregistration should freeze independent inversion implementations, global energy domains, numerical tolerances, and model variations before evaluating new outcomes. In particular, a positive DF on a finite sample must not be described as a proof of phase-space admissibility over all energies.

10. Hydrus I and observational boundaries

Hydrus I provides observational motivation and model-inspired input scales in portions of the broader UDCT program. The archived v7.13 and v7.14 internal Jeans/Hessian and phase-space stress tests are not equivalent to a direct likelihood fit to the complete individual-star line-of-sight velocity data. No such fit is demonstrated in this note. The 3D external-field problem, membership uncertainty, observational systematics, and full dynamical stability require separately defined tests.

11. Limitations and independence

All six audits share physical assumptions with the archived Step 47 field and inversion kernel. Different quadrature resolutions and optimization algorithms test numerical stability but are not independent derivations of the physical model. The scans use finite intervals and grids; extremely small differences between algorithms may exceed realistic physical accuracy. No certified interval arithmetic, full continuum proof, general nonspherical GQUMOND calculation, or parameter-free observational discrimination is claimed. The model includes fixed mass, scale and acceleration inputs; parameter-free should not be inferred from the absence of a fit in the current exercise.

12. Roadmap and preregistration separation

The present note should be published as exploratory findings first. A distinct v7.14 preregistration may subsequently state novel falsifiable questions, freeze model classes and datasets, preregister thresholds and negative controls, and require independent code paths. Binary systems, Bullet Cluster lensing and cross-scale invariance remain long-term future directions only; no universality is claimed from a shared numerical exponent.

13. Future Directions: Near-Term Tests and Long-Term Extensions

This section distinguishes prospective tests motivated by the present exploratory audit from broader research ambitions. None of the activities below is represented as a completed result or as evidence of a universal p = 1.5 threshold.

12.1 Near-term: v7.14 preregistration and confirmatory tests

• Freeze the exact QUMOND interpolation, tracer family, boundary conditions, nondimensionalization, Osipkov–Merritt inversion convention, energy domain, and numerical tolerances before new runs. Explicitly separate the central asymptotic coefficient cancellation at p = 1.5 from the finite-energy local tangency curve.

• Preregister new p values and anisotropy radii not used to locate the exploratory boundary; specify a priori success/failure thresholds, convergence criteria, negative controls, and an independent computational implementation. Retain null and contradictory outcomes.

• Test sensitivity to central regularization, quadrature order, optimization brackets, finite-energy sampling, and alternative tracer profiles. Assess global distribution-function nonnegativity over a prespecified energy domain rather than extrapolating from local minima.

• Extend the consistency checks toward stationary Jeans admissibility, three-dimensional Hessian diagnostics, and phase-space distribution-function reconstruction, distinguishing mathematical consistency from dynamical stability.

• Only after the model-level tests are frozen should observational confrontation be planned: use documented Hydrus I inputs and, in a separate stage, stellar line-of-sight velocities with selection functions, measurement uncertainties, membership modeling, and likelihood-based comparison. The present study does not perform that direct fit.

12.2 Longer-term: independent astrophysical and cross-scale tests

• Binary-star and wide-binary systems: construct a separate dynamical and external-field model and test it against suitable astrometric and kinematic data, with conventional gravity and existing MOND formulations as controls.

• Bullet Cluster and other merging galaxy clusters: develop a time-dependent, three-dimensional treatment including baryonic mass maps, gas dynamics, lensing observables, and merger geometry. A static or local p = 1.5 result alone cannot address cluster lensing offsets.

• Cross-scale and multiscale invariance: test whether any dimensionless boundary structure survives changes of mass profile, size, environment, acceleration regime, and observable. Reappearance of a similar exponent in one additional system would not by itself demonstrate universality.

• If the model-level consistency tests succeed, explore two-way field–matter back-reaction and alternative dynamical sectors; evaluate conservation laws, well-posedness, stability, and independent observational predictions before advancing a new gravity claim.

Publication sequence: exploratory findings (this note) → separately frozen v7.14 preregistration → post-registration execution and audit → later extensions only if justified by the results.

14. Conclusions

In the specified isolated spherical QUMOND tracer model, p=1.5 marks a cancellation of the leading central asymptotic second-derivative coefficient. The archived OM inversion has a separate finite-energy local tangency at ra/a approximately 0.4664667874 for p=1.5, with a smoothly varying critical curve around that exponent. Audits 01–06 improve traceability and local numerical convergence checks, but do not establish a universal QUMOND threshold, global DF positivity, stability, or observational confirmation. The scientifically defensible result is a bounded, reproducible model-specific distinction that motivates genuinely prospective preregistered tests.

References — External Scholarly Literature

1. Bekenstein, J. & Milgrom, M. (1984). Does the missing mass problem signal the breakdown of Newtonian gravity? Astrophysical Journal 286, 7–14. https://doi.org/10.1086/162570

2. Milgrom, M. (2010). Quasi-linear formulation of MOND. MNRAS 403, 886–895. https://doi.org/10.1111/j.1365-2966.2009.16184.x

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9. Binney, J. & Tremaine, S. (2008). Galactic Dynamics, 2nd ed. Princeton University Press.

References — UDCT Research Records and Provenance

The prior UDCT version sequence (v7.08–v7.13) is cited as the author's research provenance, not as independent peer-reviewed confirmation. Specific historical DOI strings should be verified against published Zenodo records before final release. Primary materials actually used here are the user-provided v7.14 Steps 01–50 archive pair and six audit ZIPs; their hashes and file listings are included in the companion manifest.

1. Paik, W. S. (2026). UDCT v7.14 exploratory Steps 01–50. User-provided archives: "7.14 Hessian 1~25.zip" and "7.14 Hessian 26~50.zip".

2. Paik, W. S. (2026). UDCT v7.14 Exploratory Audits 01–06. User-provided archive: "7.14 Exploratory audit 1~6.zip"; executable audit subarchives.

3. Paik, W. S. (2026). UDCT v7.14 Exploratory Findings: The p=1.5 Boundary in QUMOND. Reserved DOI 10.5281/zenodo.23695048; https://github.com/wspaik5/UDCT_v7.14_Exploratory_p1.5_Boundary (publication pending).

4. Paik, W. S. (2026). UDCT v7.08–v7.13 preregistration, post-registration and GQUMOND prior-art correction research records. Verify individual DOI records against Zenodo prior to public citation.

Source provenance and reproducibility statement

The companion package includes exact SHA-256 hashes of the three source ZIPs, the six audit ZIPs, and the Audit 06 tables, together with the unmodified Audit 06 executable archive. The figures in this note were generated from the archived Audit 06 CSV values. No new independent full-physics rerun is claimed for this synthesis. Readers can reproduce the original local calculations from the archived audit scripts and their declared dependencies.

Appendices A–F: Detailed Audit Record

These appendices are newly typeset explanatory material appended after the six original pages. The original PDF pages, including their text, tables, figures, conclusions, two reference sections and original pagination, are retained without rewriting. Appendix numerical statements are transcribed from the six archived audit packages; no fresh full-physics rerun is claimed.

Notation: p = tracer-shape exponent; a = model scale; r_a = Osipkov–Merritt anisotropy radius; r/a in the inversion calculations is an energy label, not an observed orbital radius; DF = distribution function. Numerical precision in printed tables is not a certified error bound.

Audit archive manifest

Audit	Original ZIP	SHA-256 (prefix)
01	UDCT_v7.14_Exploratory_Audit_01.zip	ca0343666bb7cf502374...
02	UDCT_v7.14_Exploratory_Audit_02.zip	0dd6e73ea6ef93dd7134...
03	UDCT_v7.14_Exploratory_Audit_03.zip	1d9bbc7ac9938812b7dd...
04	UDCT_v7.14_Exploratory_Audit_04.zip	c99c748c500fbce3072f...
05	UDCT_v7.14_Exploratory_Audit_05.zip	e86c5c45c9d5e8c794f1...
06	UDCT_v7_14_Exploratory_Audit_06.zip	71178608a1d634acfd12...

A. Audit 01 — Asymptotic and independent-quadrature verification

A.1 Objective and inputs

Audit 01 compares the Step 40 central inverse-series argument with a symbolic derivative check and selected Step 50 finite-energy Osipkov–Merritt inversions with an alternative adaptive quadrature. The archive also includes unchanged original Step 40 and Step 50 scripts and results. Both comparisons are exploratory, and the quadrature calculation retains the source gravitational and inversion kernels.

A.2 Asymptotic argument

For the Plummer gravitating mass, g_N(r)=GMr/(r²+a²)^(3/2), so g_N is proportional to r at small r. The isolated simple-QUMOND acceleration g=g_N/2+sqrt(g_N²/4+a_0 g_N) is proportional to r^(1/2) at leading order. The central potential difference therefore scales as Psi proportional to r^(3/2). The tracer rho_p(r)=[1+(r/a)^p]^(-5/p) has rho_p=1-(5/p)(r/a)^p+...; consequently rho_p(Psi)=1-C Psi^(2p/3)+..., C>0. The leading second derivative is -C alpha(alpha-1)Psi^(alpha-2), alpha=2p/3, and its coefficient vanishes at p=1.5.

This is cancellation of one asymptotic term. Higher-order terms matter exactly at p=1.5; the full finite-energy DF need not vanish. This distinction is the primary mathematical caution motivating the later audits.

A.3 Adaptive quadrature and evidence

Diagnostic	Archived value
Integration	SciPy adaptive quad with t=sqrt(log(r/r0)) substitution
Evaluated samples	15
Negative samples	1

Diagnostic	Archived value
Integration warnings	0

Selected original adaptive-quadrature table rows (see companion CSV for all columns and full precision):

p	ra_over_a	energy_label_r_over_a	df
1.5	0.46	0.474097552832229	-5.757231477049456
1.5	0.466466787396	0.01	0.0185117053081338
1.5	0.466466787396	0.2	0.0015030157810879
1.5	0.466466787396	0.35	0.0001692605929428
1.5	0.466466787396	0.43	1.59098149014528e-
1.5	0.466466787396	0.47	1.1909331261691536
1.5	0.466466787396	0.474097552832229	1.5274142900158996

A.4 Reproduction and limits

Run symbolic_step40_audit.py and independent_quad_audit.py after installing the included requirements. The adaptive results cover 15 selected points only. They do not certify a continuum sign, and agreement between integration methods cannot rule out shared errors in the model or inversion formula.

B. Audit 02 — Extended finite-energy sign scan

B.1 Question and numerical protocol

At the p=1.5 local tangency identified in the exploratory Step 50 calculation, does a finite energy-label scan resolve a negative DF? Audit 02 fixes p=1.5 and r_a/a=0.466466787396, evaluates 61 energy-label points on r/a in [0.01,30], and performs bounded local minimization on [0.36,0.60]. The included step47_reference.py is the common physical kernel.

B.2 Archived numerical outcomes

Quantity	Result
Tracer exponent p	1.5
OM radius r_a/a	0.466466787396
Energy-label domain	[0.01, 30.0]
Grid samples	61
Strictly negative grid values	1
Minimum sampled DF	-1.429771859502e-16
Minimum grid energy label	0.474097552832
Locally optimized DF	-1.422916815544e-16
Locally optimized energy label	0.474097563301
Max absolute N96 vs N192 difference	2.300317922255e-12

B.3 Interpretation

The negative minimum is approximately 1.4e-16 in the archive's DF normalization, i.e. numerically indistinguishable from a zero tangency at the reported precision without a certified error interval. The single negative grid value is not evidence of a robust negative lobe. Conversely, a finite scan cannot prove that the DF is nonnegative at all energies.

B.4 Reproduction and limitations

Install requirements and run audit02.py. It regenerates finite_domain_scan.csv and audit02_results.json. The grid is finite, the minimization is local, and the inversion kernel is shared with the earlier step. No individual-star Hydrus I likelihood or external-field simulation is performed.

C. Audit 03 — Critical anisotropy curve near p=1.5

C.1 Purpose and local critical-radius construction

Audit 03 asks whether the local zero-DF tangency r_a/a is stationary or discontinuous at p=1.5. It uses the affine Osipkov–Merritt decomposition $f(Q,r_a)=f_{iso}(Q)+B(Q)/(r_a/a)^2$. Where $f_{iso}>0$ and $B<0$, the candidate local critical radius is $\sqrt{-B/f_{iso}}$. The algorithm maximizes this candidate over the restricted energy-label interval r/a in [0.25,0.75].

C.2 Six-point archived curve

p	Critical r_a/a (N192 at N96 optimum)	Energy label r/a
1.490	0.462739644607	0.476369776
1.495	0.464599468922	0.475243086
1.499	0.466092716172	0.474328250
1.500	0.466466787396	0.474097647
1.501	0.466841165090	0.473866280
1.505	0.468341767989	0.472933131

C.3 Quadrature-order comparison

The archived 96-node optimization is followed by a 192-node evaluation at the same energy label; this is not a full re-optimization at 192 nodes. Absolute N192–N96 critical-radius differences in the six rows range from roughly 5e-14 to 1e-12. These small differences indicate numerical stability of this specific calculation but not physical accuracy to twelve decimal places.

C.4 Interpretation and limits

The sampled local critical radius rises from about 0.462739645 at p=1.490 to 0.468341768 at p=1.505. The local tangency energy label shifts downward over the same range. This is evidence for a smoothly varying finite-energy local boundary in the specified model, distinct from the p=1.5 central asymptotic coefficient cancellation. It does not establish a global positivity boundary, analyticity, or applicability to other tracer families.

Reproduce with audit03.py, which writes critical_vs_p.csv and results.json. The search bracket, quadrature order and physical kernel are fixed as archived.

D. Audit 04 — Parameter-grid sign stress test

D.1 Protocol

Audit 04 evaluates 3 tracer exponents p={1.499,1.500,1.501}, 4 OM settings r_a/a={0.45,0.466466787396,0.50,infinity}, and 18 energy-label values r/a between 0.01 and 30. This is 216 DF evaluations. The code uses 64-point Gauss–Legendre quadrature and the exact affine decomposition from the shared Step 47 kernel. A sample is counted negative only when f<-1e-10.

D.2 All twelve archived parameter combinations

p	r_a/a	Negative / 18	Minimum sampled DF
1.499	0.45	7	-1.5432351e-04
1.499	0.466466787396	0	4.9534452e-07
1.499	0.5	0	4.3116419e-07
1.499	inf	0	2.5942292e-10
1.500	0.45	7	-1.5818197e-04
1.500	0.466466787396	0	6.6530982e-15
1.500	0.5	0	4.3122411e-07
1.500	inf	0	2.5945275e-10
1.501	0.45	7	-1.6204600e-04
1.501	0.466466787396	3	-3.2653149e-06
1.501	0.5	0	4.3128381e-07
1.501	inf	0	2.5948246e-10

D.3 Interpretation

At r_a/a=0.45, seven sampled values are negative for each of the three p values. At the p=1.5 reference tangency r_a/a=0.466466787396, no values cross the negative threshold for p=1.499 or 1.500, whereas three do for p=1.501. At r_a/a=0.50 and in the isotropic limit, none of the 18 samples cross the threshold for any of the three p values. These are finite-grid counts, not continuum sign classifications.

D.4 Reproduction and limitations

Run audit04.py to regenerate finite_domain_sign_scan.csv and summary.json. The threshold -1e-10 is a numerical classification convention, not a physical transition criterion. Near-zero results require higher quadrature order and local minimization, addressed in Audit 05. No independent model implementation or observational likelihood is involved.

E. Audit 05 — High-resolution negative-lobe convergence

E.1 Fixed model and search

Audit 05 holds p=1.501 and r_a/a=0.466466787396 fixed, then minimizes the DF over the finite energy-label interval r/a in [0.25,0.75]. It compares Gauss–Legendre quadrature with N=96, 192 and 384 nodes. It also performs a separate local zero-tangency search for r_a/a in [0.45,0.50] at 192 nodes.

E.2 Archived convergence results

Nodes	Minimum DF	Minimum energy label r/a
96	-3.2713868467321e-06	0.473167428686
192	-3.2713868390085e-06	0.473167428914
384	-3.2713868336334e-06	0.473167421101

The minimum stays near -3.27138684e-6 across the three quadrature orders, substantially larger in magnitude than the near-zero rounding-scale residual of Audit 02. The 192-node local zero-tangency result at p=1.501 is r_a/a=0.466841165090 at energy label r/a=0.473866183640.

E.3 Scientific significance

The convergent negative lobe at p=1.501 and the p=1.5 reference anisotropy radius is a more robust local numerical sign result than a single nearly-zero grid point. It supports movement of the local critical anisotropy radius with p. It is not evidence of a singular universal transition at p=1.5: the tested r_a/a was deliberately held at the p=1.5 local tangency.

E.4 Remaining uncertainties

Quadrature convergence and optimizer convergence are distinct. The reported digits do not provide certified total errors. The local search interval excludes other energy regions, and a shared inversion kernel could introduce common systematic error. Run audit05.py to regenerate minima_convergence.csv and summary.json.

F. Audit 06 — Optimizer agreement and local derivatives

F.1 Purpose and methods

Audit 06 cross-checks the local tangency maximization using bounded scalar optimization and golden-section search on the shared numerical function. The optimizer comparison uses energy labels r/a in [0.43,0.52] at p=1.499, 1.500 and 1.501. The finite-difference derivative estimates at p=1.5 use the archived Audit 03 curve at two step sizes; they are not analytic derivatives.

F.2 Optimizer comparison

p	Bounded r_a/a	Golden r_a/a	Abs. difference
1.499	0.466092716172	0.466092716172	0.000e+00
1.500	0.466466787397	0.466466787397	5.551e-17
1.501	0.466841165091	0.466841165091	0.000e+00

F.3 Finite-difference slope and curvature

h	Estimated slope	Estimated curvature
0.001	0.374224460	0.306469978
0.005	0.374229907	0.306484746

The largest reported optimizer discrepancy in r_a/a is 5.55e-17, reflecting agreement in floating-point evaluation of the same objective. The slope estimates near 0.37422 and curvature near 0.30648 are stable at the two tested finite-difference spacings within their reported variation. Two spacings alone cannot certify smoothness or analyticity beyond the sampled region.

F.4 Reproduction, provenance and interpretation

Run audit06.py with step47_reference.py and the archived Audit 03 critical_vs_p.csv. Outputs are optimizer_comparison.csv, curve_derivatives.csv, results.json and run.log. This is an algorithmic numerical cross-check, not an independent physical inversion. The local tangency curve appears smooth in the tested neighborhood, so the central p=1.5 cancellation and finite-energy OM tangency must remain conceptually separate.

Appendix G — Reproducibility map and audit boundaries

This appendix adds an explicit source map without modifying the six-page main article. All original six ZIP packages are preserved separately. The accompanying appendix ZIP contains these six original audit archives without recompression of their internal members, a SHA-256 inventory, and this complete merged PDF.

Audit	Primary script	Primary output
01	symbolic_step40_audit.py; independent_quad_audit.py	adaptive_quad_results.csv; adaptive_quad_summary.json
02	audit02.py	finite_domain_scan.csv; audit02_results.json
03	audit03.py	critical_vs_p.csv; results.json
04	audit04.py	finite_domain_sign_scan.csv; summary.json
05	audit05.py	minima_convergence.csv; summary.json
06	audit06.py	optimizer_comparison.csv; curve_derivatives.csv; results.json

The six audit archives provide evidence of exploratory numerical checks under the specified spherical QUMOND and Osipkov–Merritt assumptions. They do not establish global phase-space admissibility, dynamical stability, a new gravity law, or a likelihood fit to individual Hydrus I stellar velocities. The proposed v7.14 preregistration must separately freeze new tests and decision thresholds before confirmatory execution.